

Safety Data Sheet

According to Notification of Thailand Ministry of Industry B.E. 2555 (2012)



PETRONAS GEAR MEP 460

Version 0.0

Release Date: 2019/08/14

1. PRODUCT AND COMPANY IDENTIFICATION

Product name : PETRONAS GEAR MEP 460
Other means of identification : None
Recommended use : Lubricating oil
Limitation of use : This product should not be used for purposes other than those specified without the advice of an expert
Supplier : PETRONAS International Marketing (Thailand) Co., Ltd
Registered corporate address : 388 Exchange Tower, Unit Nos. 4101-4104 41st floor,
Sukhumvit Road, Kweang Klongtoey, Khet Klongtoey,
Bangkok 10110
Emergency phone numbers : +662-663 5881-5

2. HAZARDS IDENTIFICATION

GHS Classification

Not a hazardous substance or mixture.

GHS Label element

Not a hazardous substance or mixture.

Other hazards which do not result in classification

No information available.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Mixture

Hazardous components

Chemical Name	CAS-No.	Concentration (%)
Paraffinic Distillate, Heavy, Hydrotreated	64742-54-7	>= 90 - <= 100
Long chain alkyl amine	68955-53-3	>= 0.1 - < 1

4. FIRST AID MEASURES

If inhaled : If unconscious, place in recovery position and seek medical advice.
If symptoms persist, call a medical doctor.
In case of skin contact : If on skin, rinse well with water.
In case of eye contact : Flush eyes with water as a precaution.
Remove contact lenses.
Protect unharmed eye.
Keep eye wide open while rinsing.
If eye irritation persists, consult a specialist.
If swallowed : Keep respiratory tract clear.
Do NOT give milk or alcoholic beverages.
Never give anything by mouth to an unconscious person.
If symptoms persist, call a medical doctor.
General advice : Move out of dangerous area.
Show this safety data sheet to the doctor in attendance.
Do not leave the victim unattended.

Most important symptoms and effects, both acute and delayed : No information available.

5. FIREFIGHTING MEASURES

Unsuitable media extinguishing : High volume water jet

Specific methods extinguishing : Standard procedure for chemical fires.
Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

Special protective equipment for firefighters : Wear self-contained breathing apparatus for firefighting if necessary.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures : Use personal protective equipment.

Environmental precautions : Prevent product from entering drains.
Prevent further leakage or spillage if safe to do so.

Methods and materials for containment and cleaning up : Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust).
Keep in suitable, closed containers for disposal.

7. HANDLING AND STORAGE

Advice on protection against fire and explosion : Normal measures for preventive fire protection.

Advice on safe handling : Do not breathe vapours/dust.
Avoid exposure - obtain special instructions before use. For personal protection see section 8. Smoking, eating and drinking should be prohibited in the application area.
Dispose of rinse water in accordance with local and national regulations.
Persons susceptible to skin sensitisation problems or asthma, allergies, chronic or recurrent respiratory disease should not be employed in any process in which this mixture is being used.
Avoid contact with skin and eyes.

Conditions for safe storage : Keep container tightly closed in a dry and well-ventilated place.
Electrical installations / working materials must comply with the technological safety standards.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Components with workplace control parameters

Components	CAS-No.	Value type (Form of exposure)	Control parameters / Permissible concentration	Basis
Paraffinic Distillate, Heavy, Hydrotreated	64742-54-7	TWA (Inhalable fraction)	5 mg/m ³	ACGIH
	Further information: Upper Respiratory Tract irritation, Not classifiable as a human carcinogen			

Personal protective equipment

Hand protection

Remarks : The suitability for a specific workplace should be discussed with the producers of the protective gloves.

Eye protection : Eye wash bottle with pure water.
Tightly fitting safety goggles.

Skin and body protection : Impervious clothing.
Choose body protection according to the amount and concentration of the dangerous substance at the work place.

Hygiene measures : Wash hands before breaks and at the end of workday.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance	: Liquid
Colour	: No data available
Odour	: Not relevant
Odour Threshold	: Not relevant
pH	: Not applicable
	: Not determined
Flash point	: 252 °C Method: Cleveland open cup
Evaporation rate	: Not applicable
Upper explosion limit	: Not applicable
Lower explosion limit	: Not applicable
Relative vapour density	: Not applicable
Relative density	: Not applicable
Density	: 0.904 g/cm ³ (15 °C)
Solubility(ies)	
Water solubility	: Not applicable
Solubility in other solvents	: Not applicable
Partition coefficient: n-octanol/water	: Not applicable
Decomposition temperature	: Not applicable
Viscosity, dynamic	: Not applicable
Viscosity, kinematic	: 460 mm ² /s (40 °C) 30.4 mm ² /s (100 °C)

10. STABILITY AND REACTIVITY

Reactivity	: Hazardous polymerisation does not occur.
Chemical stability	: Stable under normal conditions.
Possibility of hazardous reactions	: No dangerous reaction known under conditions of normal use.
Conditions to avoid	: Heat, sparks, flame and build-up of static electricity.
Incompatible materials	: Strong oxidizing agents.
Hazardous decomposition products	: Fumes, smoke, carbon monoxide

11. TOXICOLOGICAL INFORMATION**Acute toxicity****Product:**

Acute inhalation toxicity	: Acute toxicity estimate: > 40 mg/l Exposure time: 4 h Test atmosphere: vapour Method: Calculation method
Acute dermal toxicity	: Acute toxicity estimate: > 5,000 mg/kg Method: Calculation method

Components:**Paraffinic Distillate, Heavy, Hydrotreated:**

Acute oral toxicity	: LD50 (Rat): > 5,000 mg/kg
Acute inhalation toxicity	: LC50 (Rat): > 5.53 mg/l
Acute dermal toxicity	: LD50 (Rabbit): > 2,000 mg/kg

Long chain alkyl amine:

Acute oral toxicity	: LD50 (Rat): 612 mg/kg
Acute inhalation toxicity	: Assessment: The component/mixture is highly toxic after short term inhalation.
Acute dermal toxicity	: LD50 (Rat): 251 mg/kg

Skin corrosion/irritation**Product:**

Remarks	: May cause skin irritation and/or dermatitis.
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Components:**Paraffinic Distillate, Heavy, Hydrotreated:**

Species	: Rabbit
Result	: No skin irritation

Long chain alkyl amine:

Result	: Corrosive after 3 minutes to 1 hour of exposure
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Serious eye damage/eye irritation**Product:**

Remarks	: Vapours may cause irritation to the eyes, respiratory system and the skin.
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Components:**Paraffinic Distillate, Heavy, Hydrotreated:**

Species	: Rabbit
Result	: No eye irritation

Long chain alkyl amine:

Result : No eye irritation

Respiratory or skin sensitisation**Components:****Paraffinic Distillate, Heavy, Hydrotreated:**

Exposure routes : Skin contact
Result : No evidence of skin sensitization.
Exposure routes : Inhalation
Remarks : No data available

Long chain alkyl amine:

Result : The product is a skin sensitizer, sub-category 1A.

Germ cell mutagenicity**Components:****Paraffinic Distillate, Heavy, Hydrotreated:**

Germ cell mutagenicity - : No data available
Assessment

Carcinogenicity**Components:****Paraffinic Distillate, Heavy, Hydrotreated:**

Carcinogenicity - Assessment : Contains less than 3 % DMSO extract as measured by IP 346
'Determination of polycyclic aromatics in unused lubricating
base oils and asphaltene free petroleum fractions — Dimethyl
sulphoxide extraction refractive index method'

Reproductive toxicity**Components:****Paraffinic Distillate, Heavy, Hydrotreated:**

Reproductive toxicity - : Not suspected to be a reproductive toxicant.
Assessment

STOT - single exposure**Components:****Paraffinic Distillate, Heavy, Hydrotreated:**

Assessment : No specific organ toxicity following single exposure.

STOT - repeated exposure**Components:****Paraffinic Distillate, Heavy, Hydrotreated:**

Assessment : No specific organ toxicity following repeated exposure.

12. ECOLOGICAL INFORMATION**Ecotoxicity****Components:****Paraffinic Distillate, Heavy, Hydrotreated:**

- Toxicity to fish : LL50 (Pimephales promelas (fathead minnow)): > 100 mg/l
Exposure time: 96 h
- Toxicity to daphnia and other aquatic invertebrates : Remarks: No data available
- Toxicity to algae : NOEL (Pseudokirchneriella subcapitata (green algae)): >= 100 mg/l
Exposure time: 72 h
- Toxicity to fish (Chronic toxicity) : NOELR (Oncorhynchus mykiss (rainbow trout)): >= 1,000 mg/l
Exposure time: 14 d
- Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : NOEL (Daphnia magna (Water flea)): 10 mg/l
Exposure time: 21 d
- Toxicity to microorganisms : NOEL: > 1.93 mg/l
Exposure time: 4 d

Long chain alkyl amine:

- Toxicity to fish : LC50 (Oncorhynchus mykiss (rainbow trout)): 1.3 mg/l
Exposure time: 96 h
- Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): 0.24 - 6 mg/l
Exposure time: 48 h
- Toxicity to algae : EC50 (Pseudokirchneriella subcapitata (green algae)): 0.44 mg/l
Exposure time: 72 h
- Toxicity to fish (Chronic toxicity) : NOEC (Oncorhynchus mykiss (rainbow trout)): 0.078 mg/l
Exposure time: 96 h
- Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : Remarks: No data available
- Toxicity to microorganisms : EC50 (Activated sludge): 63.5 mg/l
Exposure time: 30 min

Persistence and degradability**Components:****Paraffinic Distillate, Heavy, Hydrotreated:**

- Biodegradability : Exposure time: 28 d
Result: Not readily biodegradable. Method: OECD Test Guideline 301

Long chain alkyl amine:

- Biodegradability : Result: Inherently biodegradable.

Bioaccumulative potential**Components:****Paraffinic Distillate, Heavy, Hydrotreated:**

- Bioaccumulation : Remarks: No data available
- Partition coefficient: : Remarks: No data available
- n-octanol/water:

Long chain alkyl amine:

Bioaccumulation : Remarks: No data available
Partition coefficient: : log Pow: 2.9 (23 °C)
n-octanol/water

Mobility in soil**Components:****Paraffinic Distillate, Heavy, Hydrotreated:**

Mobility : Medium: Soil
Remarks: No data available

Other adverse effects**Product:**

Additional ecological : No data available
information

Components:**Paraffinic Distillate, Heavy, Hydrotreated:**

Additional ecological : No data available
information

Long chain alkyl amine: No data available

13. DISPOSAL CONSIDERATIONS**Disposal methods**

Waste from residues : Do not dispose of waste into sewer.
Do not contaminate ponds, waterways or ditches with chemical
or used container.
Send to a licensed waste management company.

Contaminated packaging : Empty remaining contents.
Dispose of as unused product.
Do not re-use empty containers.

14. TRANSPORT INFORMATION**International Regulation****UNRTDG**

Not regulated as a dangerous good

IATA-DGR

Not regulated as a dangerous good

IMDG-Code

Not regulated as a dangerous good

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

15. REGULATORY INFORMATION**Safety, health and environmental regulations/legislation specific for the substance or mixture****The components of this product are reported in the following inventories:**

CH INV	: Not in compliance with the inventory
TSCA	: Not On TSCA Inventory
DSL	: This product contains the following components that are not on the Canadian DSL nor NDSL. Block copolymer of ethylene oxide and propylene oxide
AICS	: Not in compliance with the inventory
NZIoC	: Not in compliance with the inventory
ENCS	: Not in compliance with the inventory
ISHL	: Not in compliance with the inventory
KECI	: Not in compliance with the inventory
PICCS	: Not in compliance with the inventory
IECSC	: Not in compliance with the inventory

16. OTHER INFORMATION

Revision Date : 2019/08/14

Full text of other abbreviations

(Q)SAR	-	(Quantitative) Structure Activity Relationship
ACGIH	-	American Conference of Governmental Industrial Hygienists
AICS	-	Australian Inventory of Chemical Substances
ANTT	-	National Agency for Transport by Land of Brazil
ASTM	-	American Society for the Testing of Materials
bw	-	Body weight
CCHC	-	Chemicals Classification and Hazard Communication
CMR	-	Carcinogen, Mutagen or Reproductive Toxicant
CPR	-	Controlled Products Regulations
DIN	-	Standard of the German Institute for Standardisation
DSL	-	Domestic Substances List (Canada)
ECx	-	Concentration associated with x% response
ELx	-	Loading rate associated with x% response
EmS	-	Emergency Schedule
ENCS	-	Existing and New Chemical Substances (Japan)
ErCx	-	Concentration associated with x% growth rate response
ERG	-	Emergency Response Guide
GHS	-	Globally Harmonized System
GLP	-	Good Laboratory Practice
IARC	-	International Agency for Research on Cancer
IATA	-	International Air Transport Association
IBC	-	International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk
IC50	-	Half maximal inhibitory concentration
ICAO	-	International Civil Aviation Organization
ICOP	-	Industry Code of Practice on Chemicals Classification and Hazard Communication
IECSC	-	Inventory of Existing Chemical Substances in China
IMDG	-	International Maritime Dangerous Goods
IMO	-	International Maritime Organization
ISHL	-	Industrial Safety and Health Law (Japan)

ISO	-	International Organisation for Standardization
KECI	-	Korea Existing Chemicals Inventory
LC50	-	Lethal Concentration to 50 % of a test population
LD50	-	Lethal Dose to 50% of a test population (Median Lethal Dose)
MARPOL	-	International Convention for the Prevention of Pollution from Ships
MY PEL	-	Malaysian Permissible Exposure Limit
n.o.s.	-	Not Otherwise Specified
Nch	-	Chilean Norm
NITE	-	National Institute of Technology and Evaluation
NO(A)EC	-	No Observed (Adverse) Effect Concentration
NO(A)EL	-	No Observed (Adverse) Effect Level
NOELR	-	No Observable Effect Loading Rate
NOM	-	Official Mexican Norm
NTP	-	National Toxicology Program
NZIoC	-	New Zealand Inventory of Chemicals
OCSPP	-	Office of Chemical Safety and Pollution Prevention
OECD	-	Organization for Economic Co-operation and Development
PBT	-	Persistent, Bioaccumulative and Toxic
PICCS	-	Philippines Inventory of Chemicals and Chemical Substances
REACH	-	Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals
SADT	-	Self-Accelerating Decomposition Temperature
SDS	-	Safety Data Sheet
STEL	-	Short Term Exposure Limit
TCSI	-	Taiwan Chemical Substance Inventory
TDG	-	Transportation of Dangerous Goods
TSCA	-	Toxic Substances Control Act (United States)
TWA	-	Time Weighted Average
UN	-	United Nations
UNRTDG	-	United Nations Recommendations on the Transport of Dangerous Goods
UVCB	-	Unknown or Variable Composition, Complex Reaction Products and Biological Materials
vPvB	-	Very Persistent and Very Bioaccumulative
WHMIS	-	Workplace Hazardous Materials Information System

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